SigmaTau.jl: time-and-frequency stability analysis in Julia

Every estimator family, per-τ noise ID, EDF and χ^2 intervals in one call · matches Stable32 and allantools · modified-total kernels 1,600 to 4,200× faster · new Monte-Carlo-calibrated MHTOTDEV

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One API: every estimator, noise ID, EDF, χ^2 bounds

Estimators: ADEV/MDEV/TDEV; HDEV/MHDEV/HTDEV; TOTDEV, MTOTDEV, TTOTDEV, HTOTDEV, MHTOTDEV; Thêo1/ThêoBR/ThêoH; MTIE, TIE rms, PDEV; DADEV/DHDEV; S_y, S_x, $\mathscr{L}(f)$; calibrated power-law synthesis.

Per τ: σ, noise type (lag-1; B1/R(n) fallback), Greenhall–Riley EDF, 68.3 % χ^2 bounds (switchable). Total-family bias correction default on, switchable.

Input: phase or frequency, any τ grid; read/detrend/gap-fill/outlier tools; plain-text round-trip.

Real-time, multi-clock: ADEV/MDEV/HDEV/MHDEV in O(1) per sample per τ; ring buffer $\le 4 \cdot m_{max} + 1$ samples; $\le 2 \times 10^{-11}$ relative to batch kernels on samples seen so far (Dobrogowski–Kasznia 2007, extended to third differences). N-cornered hat, N ≥ 3.

```
p = read_phase("clock.DAT"; tau0 = 1.0)
r = adev(p)               # τ grid, σ_y, noise type, EDF, CI
suite = stability(p; devs = (:adev, :mdev, :hdev, :mhtotdev))
```

	Stable32	allantools	SigmaTau
Platform	Windows GUI, closed	Python library	Julia library + desktop app
χ^2 intervals with per-τ noise ID	yes	manual	every call
Total-family bias correction	TOTDEV, HTOTDEV	none	all five, switchable
Modified Hadamard Class (MHDEV, HTDEV, MHTOTDEV)	no	no	yes
Thêo1 / ThêoBR / ThêoH	yes	Thêo1 only	yes
Dynamic (time-resolved) deviation	yes	no	yes
Real-time accumulators	no	ADEV, HDEV, TDEV	ADEV, MDEV, HDEV, MHDEV
N-cornered hat	three clocks	three clocks	N ≥ 3
Multithreaded kernels	n/a	no	yes

Deviation, noise type and χ^2 interval at every τ

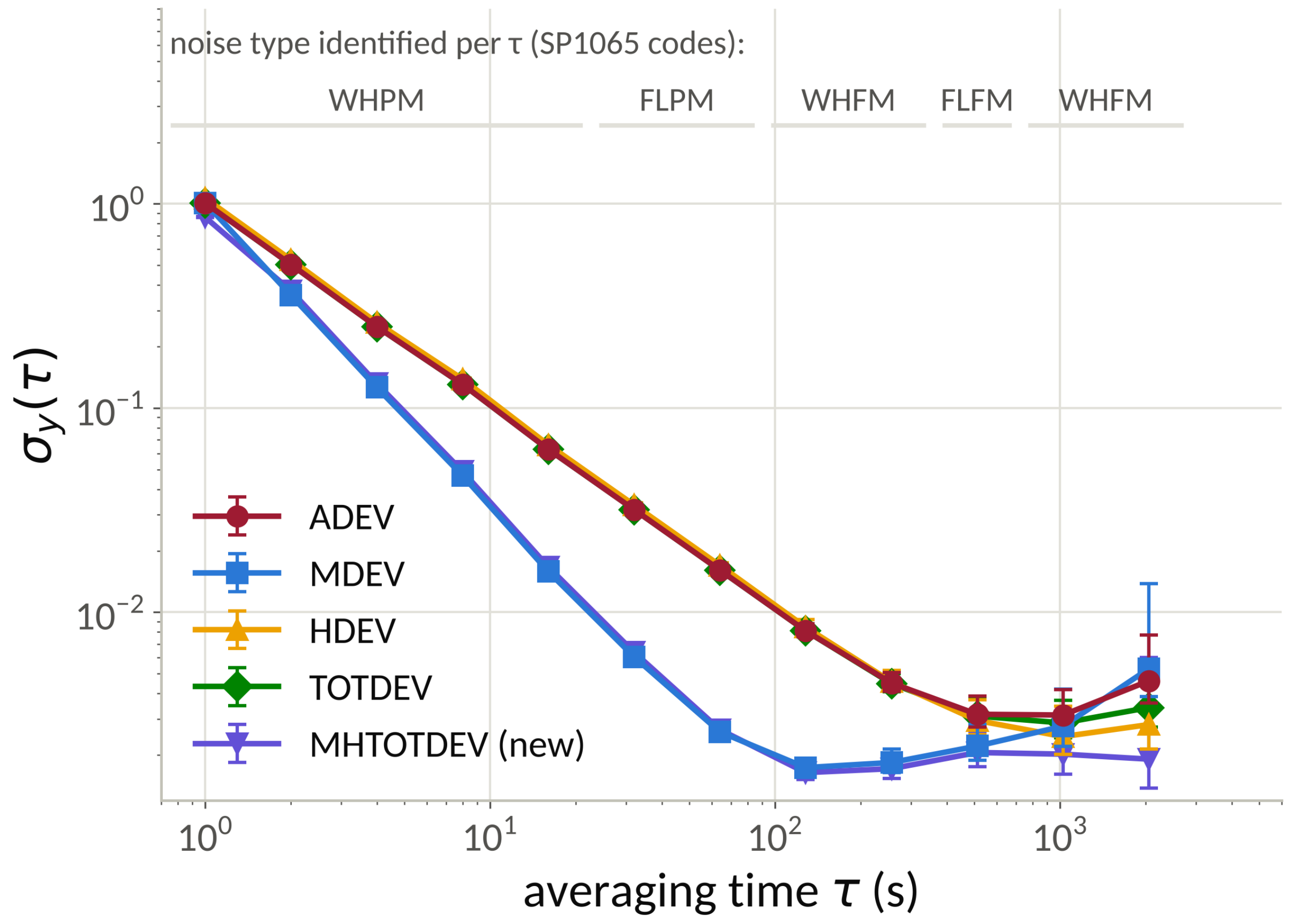


Fig. 1. Shipped 8192-point record, τ_0 = 1 s, one call: five estimators, per-τ noise type (top strip), 68.3 % χ^2 intervals. TOTDEV reaches T/2, MHTOTDEV T/4, tighter long-τ bounds.

Matches Stable32 to printed precision, allantools to machine precision

Shipped fixture: 8192-point phase record, Stable32 and allantools outputs. ADEV/MDEV/TDEV/HDEV: ≤ 0.005 % vs Stable32 (five significant figures); $\le 1 \times 10^{-13}$ vs allantools. Raw total kernels: $\le 4.4 \times 10^{-15}$ vs allantools. Confidence bounds: ≤ 0.05 % vs Stable32 at 11 of 12 τ.

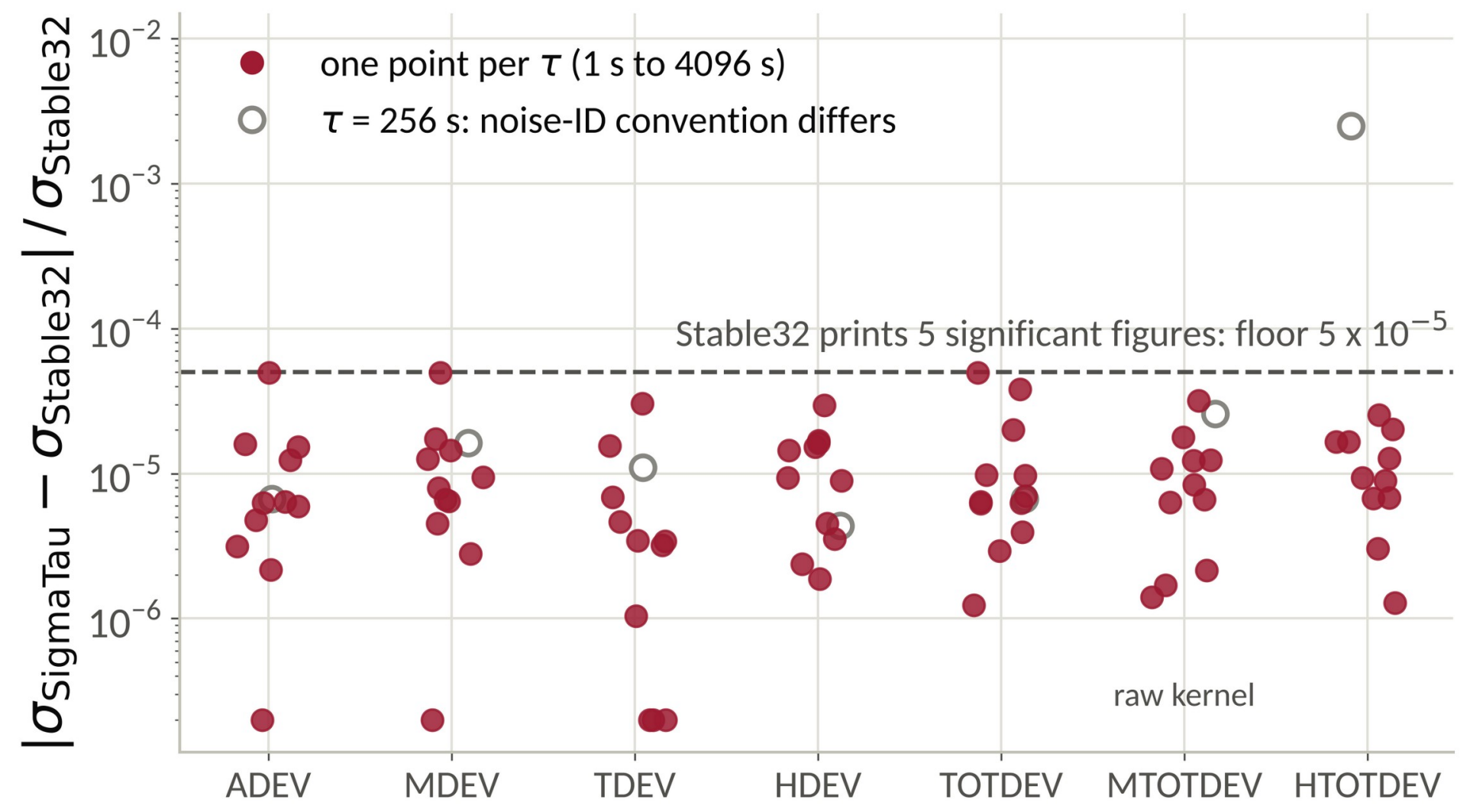


Fig. 2. Seven estimators vs Stable32, every τ. Filled: $\le 5 \times 10^{-5}$ printed-precision floor. Open: τ = 256 s, Stable32 α = +2, detrended α = 0. MTOTDEV raw.

4 × 10^5-sample modified totals: 0.46 s vs 30 min

Measured 406 763-sample phase record, kernel-only timing, same samples, τ grid and statistic; allantools single-threaded (no parallel path). MTOTDEV 0.461 s vs 1 810 s (3,926×); HTOTDEV 0.461 s vs 1 926 s (4,178×). Reflected extension streamed per thread; window sum split across threads.

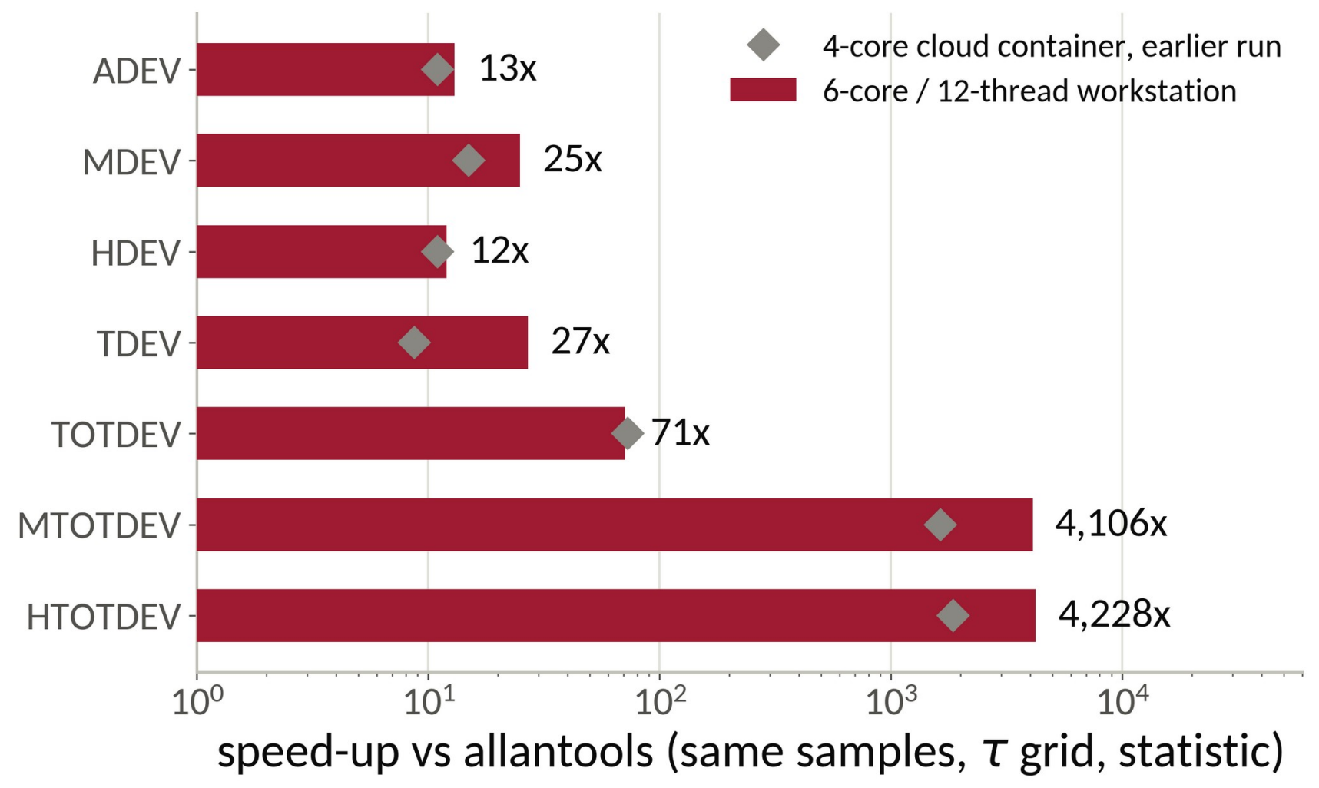
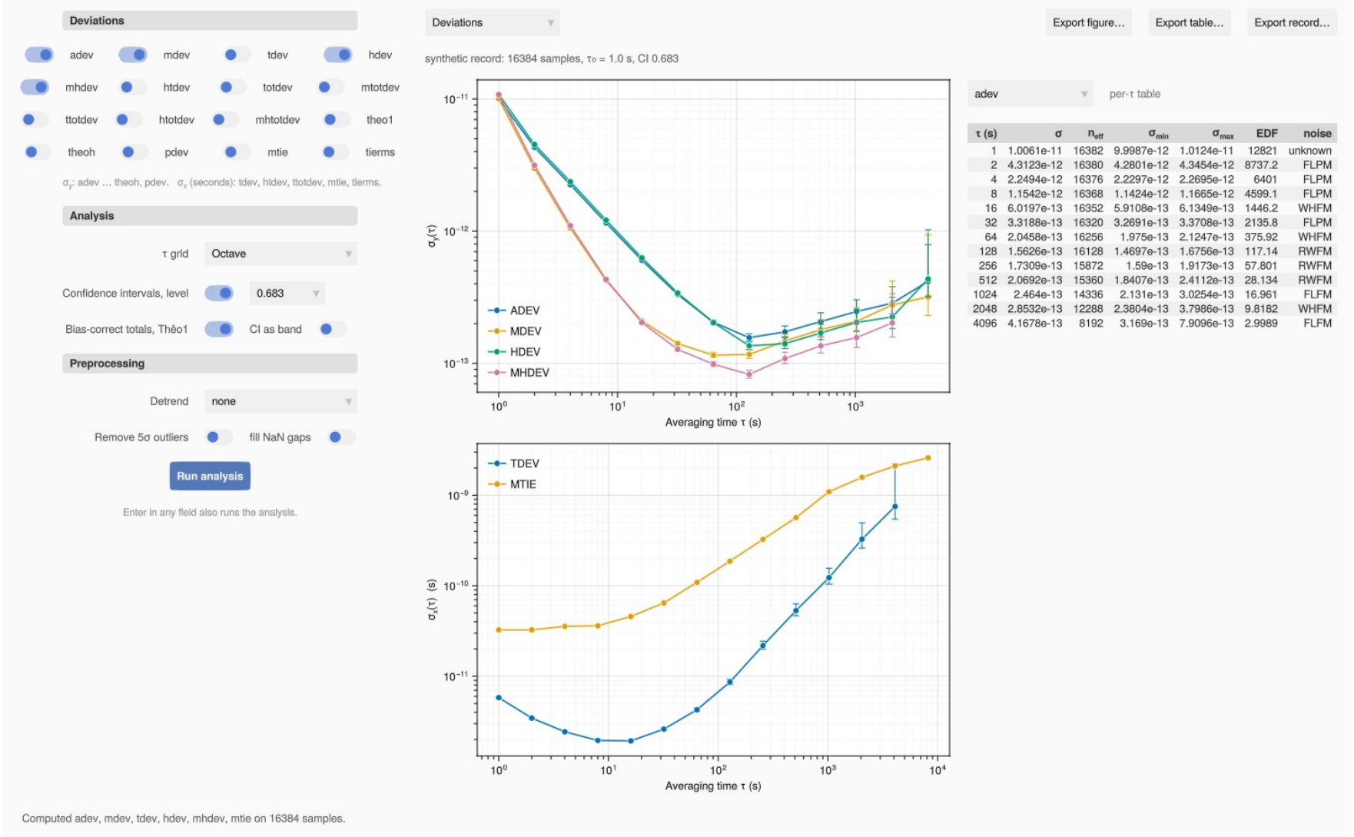


Fig. 3. Speed-up per kernel, N = 25 000 white-PM record (bench/), 12-thread workstation; diamonds: 4-core container run. Ratios portable; absolute times machine-dependent.

Desktop explorer: the whole workflow without code



σ_y: adev … theoh, pdev. σ_x (seconds): tdev, htdev, ttotdev, mtie, tierms.

τ grid: Octave

Confidence intervals, level: 0.683

Bias-correct totals, Thêo1

CI as band

Detrend: none

Remove 5σ outliers

fill NaN gaps

Enter in any field also runs the analysis.

synthetic record: 16384 samples, τ_0 = 1.0 s, CI 0.683

τ (s)	σ	n_{eff}	σ_{min}	σ_{max}	EDF	noise
1	1.0061e-11	16382	9.9987e-12	1.0124e-11	12821	unknown
2	4.3123e-12	16380	4.2801e-12	4.3454e-12	8737.2	FLPM
4	2.2494e-12	16376	2.2297e-12	2.2695e-12	6401	FLPM
8	1.1542e-12	16368	1.1424e-12	1.1665e-12	4599.1	FLPM
16	6.0197e-13	16352	5.9108e-13	6.1349e-13	1446.2	WHFM
32	3.3188e-13	16320	3.2691e-13	3.3708e-13	2135.8	FLPM
64	2.0458e-13	16256	1.975e-13	2.1247e-13	375.92	WHFM
128	1.5626e-13	16128	1.4697e-13	1.6756e-13	117.14	RWFM
256	1.7309e-13	15872	1.59e-13	1.9173e-13	57.801	RWFM
512	2.0692e-13	15360	1.8407e-13	2.4112e-13	28.134	RWFM
1024	2.464e-13	14336	2.131e-13	3.0254e-13	16.961	FLFM
2048	2.8532e-13	12288	2.3804e-13	3.7986e-13	9.8182	WHFM
4096	4.1678e-13	8192	3.169e-13	7.9096e-13	2.9989	FLFM

Computed adev, mdev, tdev, hdev, mhdev, mtie on 16384 samples.

Fig. 4. GLMakie desktop app: load or synthesize a record; preprocessing, estimators, τ grid, confidence level; σ_y and σ_x curves with χ^2 intervals, per-τ table, spectra, dynamic map; PNG/PDF/SVG and TSV export.

MHTOTDEV: new total estimator, calibrated by Monte Carlo

Monte Carlo: α ∈ {+2 … −2}, N ∈ {1024 … 32768}, R = 3000 per cell, one seeded stream per (α, N, r), fit window $\tau/\tau_0 \ge 16$. Bias B = E[MHTOTVAR]/E[MHVAR] vs MHDEV, same realization; EDF from χ^2 variance relation; bootstrap SE 2.6 %.

Kernel: Greenhall MTOTDEV/HTOTDEV scheme on 4m windows (half-mean slope removal, 3× time-reverse extension, modified third difference) after global least-squares drift removal.

PM unbiased, red FM reads high; EDF $R^2 \ge 0.9988$

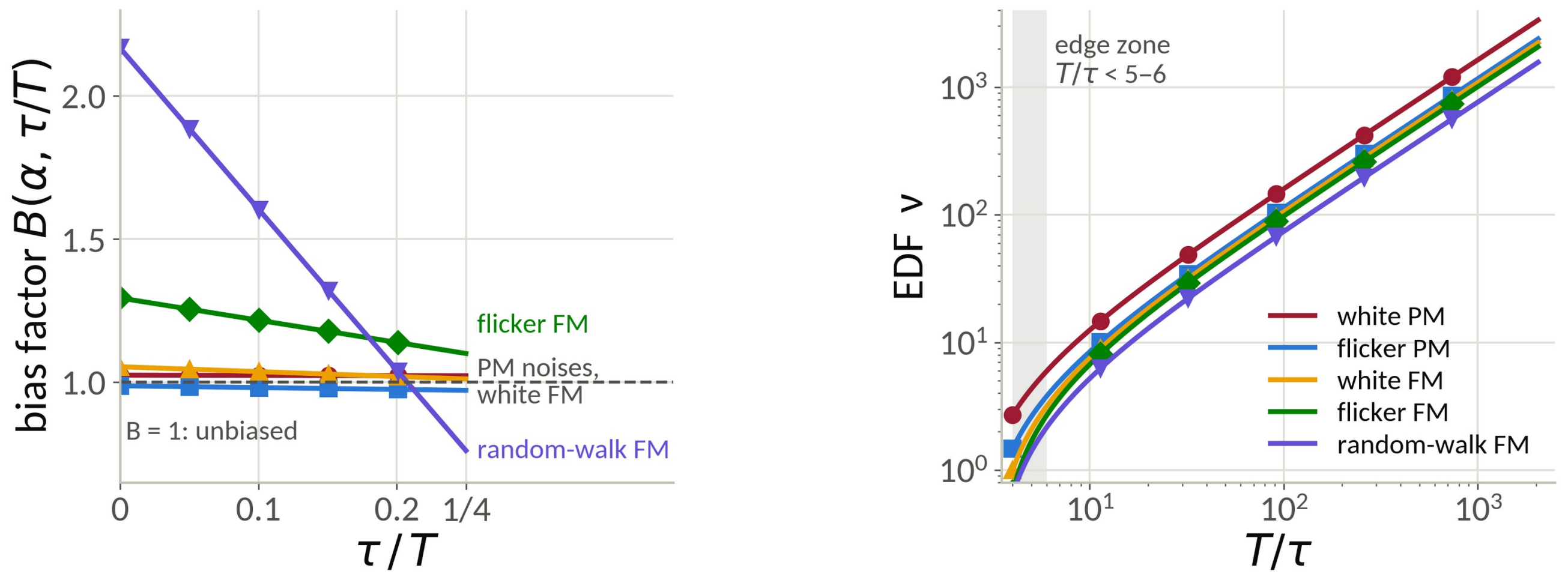


Fig. 5. Models $B = b_0 + b_1 \tau/T$, $\nu = b\, T/\tau - c$ (seed 20260531). Interval indicative only below T/τ ≈ 5–6; σ valid to τ = T/4.

α	noise	bias b_0	bias b_1	EDF b	EDF c	R^2
+2	white PM	1.0249	−0.0078	1.6342	3.8334	0.9989
+1	flicker PM	0.9879	−0.0630	1.1651	3.1934	0.9988
0	white FM	1.0545	−0.1706	1.0861	3.3451	0.9991
−1	flicker FM	1.2946	−0.7752	1.0096	3.3662	0.9992
−2	random-walk FM	2.1678	−5.6243	0.7641	2.4152	0.9988

DADEV σ(t, τ) map: change located to within one window

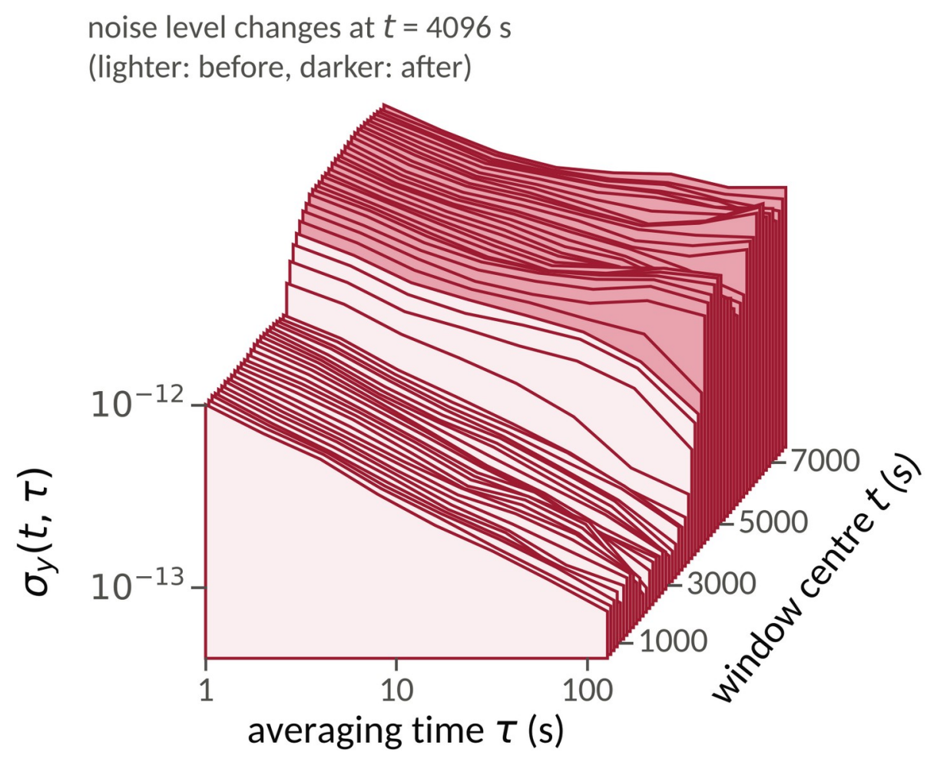


Fig. 6. Synthetic 8191-sample record, τ_0 = 1 s: white FM $\sigma_y(1\text{ s}) = 1 \times 10^{-12}$ first half, then 3×10^{-12} plus flicker FM 1×10^{-12}; window 1024, step 128.

Every result reproducible from the repository

Shipped: validation fixture with Stable32/allantools outputs; per-τ comparison tables; benchmark scripts for both libraries; Monte Carlo harness, seed, tracked fit artifact.

Next: real-time monitoring of laboratory Rb-87 clock comparisons; Monte Carlo calibration of the Thêo family and gapped records.

References

1. W. J. Riley, Handbook of Frequency Stability Analysis, NIST SP 1065 (2008).
2. IEEE Std 1139-2022, Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology.
3. C. A. Greenhall and W. J. Riley, Uncertainty of stability variances based on finite differences, Proc. 35th PTTI (2003).
4. D. A. Howe, R. L. Beard, C. A. Greenhall, F. Vernotte and W. J. Riley, A total estimator of the Hadamard function used for GPS operations, Proc. 32nd PTTI (2000).
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6. A. Dobrogowski and M. Kasznia, Real-time assessment of Allan deviation and time deviation, Proc. IEEE FCS / EFTF (2007).
7. Reference implementations compared: Stable32 (W. J. Riley) and allantools (A. E. Wallin).

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